

Tools of the Trade: Linux and SQL

Activity: Install Software in a Linux Distribution

Introduction

In this activity, I practiced how to install, uninstall, and check applications in a Linux environment using the **APT package manager**.

I used the Linux Bash shell to work with different commands and learned how `sudo` can be used when administrative permissions are required.

The applications used in this lab were:

- Nload
- tcpdump

Nload is a command-line tool that can be used to monitor network traffic and bandwidth usage, while tcpdump is a command-line tool used to capture and analyze network traffic.

Activity Scenario

In this lab, I worked as a security analyst who needed to have network security applications installed on a Linux system.

The tasks included:

1. Checking whether APT was installed.
2. Installing Nload.
3. Checking that Nload was installed.
4. Uninstalling Nload.
5. Checking that Nload was removed.
6. Installing tcpdump.
7. Listing the installed applications.
8. Reinstalling Nload.
9. Checking that Nload was installed again.

The lab used a Debian-based Linux environment where APT was already available.

Lab Environment

- **Course:** Google Cybersecurity Certificate
 - **Course:** Tools of the Trade: Linux and SQL
 - **Activity:** Install Software in a Linux Distribution
 - **Operating System:** Debian-based Linux
 - **Shell:** Bash
 - **User:** `analyst`
 - **Package Manager:** APT
-

Task 1: Check if APT is Installed

The first task was to confirm that the APT package manager was available in the Linux environment.

I entered:

```
apt
```

APT displayed information about the package manager, including its version and basic usage.

This confirmed that APT was installed and available to manage software packages on the system.

APT is commonly used on Debian-based Linux distributions to install, remove, and manage software.

Task 2: Install and Uninstall Nload

Installing Nload

The next step was to install Nload using APT.

I used:

```
sudo apt install nload
```

The `sudo` command was required because installing software requires elevated permissions.

During the installation, APT displayed information about the package and any required dependencies. I continued with the installation when prompted.

Checking the Nload Installation

After installing Nload, I checked whether it was working correctly.

I used:

```
nload -h
```

The command displayed information about Nload, including its version and usage information.

This confirmed that Nload had been installed successfully.

Uninstalling Nload

After confirming the installation, I removed Nload from the system.

I used:

```
sudo apt remove nload
```

APT asked for confirmation before removing the application, and I continued with the removal.

Checking that Nload Was Removed

After uninstalling Nload, I ran the command again:

```
nload -h
```

This time, the system returned an error because Nload was no longer installed.

The error indicated that the Nload executable could not be found.

This confirmed that the application had been successfully removed.

Task 3: Install tcpdump

The next task was to install `tcpdump`.

I used:

```
sudo apt install tcpdump
```

Tcpdump is a command-line network traffic capture tool.

It can be used by security professionals to capture and inspect network packets and is useful for network troubleshooting and security investigations.

The installation was completed using APT.

Task 4: List Installed Applications

After installing `tcpdump`, I checked the applications installed on the Linux system.

I used:

```
apt list --installed
```

This command displayed a list of installed packages.

The list was quite long because the Linux environment already had many packages installed.

I checked the list for `tcpdump` to confirm that the application had been installed.

The output included `tcpdump` as an installed package.

For example:

```
tcpdump/oldstable,now 4.9.3-1~deb10u2 amd64 [installed]
```

The exact version can be different depending on the Linux environment.

Task 5: Reinstall Nload

The final task was to install Nload again.

I used:

```
sudo apt install nload
```

After the installation finished, I checked the list of installed applications again:

```
apt list --installed
```

I searched the output for Nload.

The output showed Nload as an installed package, confirming that the application had been successfully reinstalled.

For example:

```
nload/oldstable,now 0.7.4-2+b1 amd64 [installed]
```

The exact version may be different depending on the environment.

Commands Practiced

Command	Purpose
<code>apt</code>	Check and interact with the APT package manager
<code>sudo apt install nload</code>	Install Nload
<code>nload -h</code>	Check Nload and display its usage information
<code>sudo apt remove nload</code>	Remove Nload
<code>sudo apt install tcpdump</code>	Install tcpdump
<code>apt list --installed</code>	List installed packages

What I Learned

From this activity, I learned how to manage software packages in a Linux environment using APT.

I practiced:

- Checking whether APT is available.
- Installing software using APT.
- Using `sudo` for administrative tasks.
- Checking whether an application was installed correctly.

- Removing an application.
- Confirming that an application was removed.
- Listing installed packages.
- Reinstalling an application.

I also got an introduction to two useful network-related tools, Nload and tcpdump.

Cybersecurity Connection

Software and tool management is an important part of working with Linux systems in cybersecurity.

Security analysts may need to install different tools depending on the task they are performing.

For example, **tcpdump** can be used to capture network traffic during a security investigation or troubleshooting process.

Nload can be used to monitor network traffic and bandwidth usage.

Understanding how to install and manage these types of tools gives me a better foundation for working with Linux-based security environments.

The use of `sudo` also helped me understand that some Linux operations require administrative privileges.

Practical Skills Demonstrated

- Linux Bash
 - APT package management
 - `sudo`
 - Installing Linux applications
 - Uninstalling Linux applications
 - Reinstalling applications
 - Checking installed packages
 - Basic Linux administration
 - Nload
 - tcpdump
 - Network security tool management
-

Learning Outcome

This activity gave me practical experience with managing applications in a Linux environment.

I learned how to use APT to install and remove software and how to check whether an application is installed correctly.

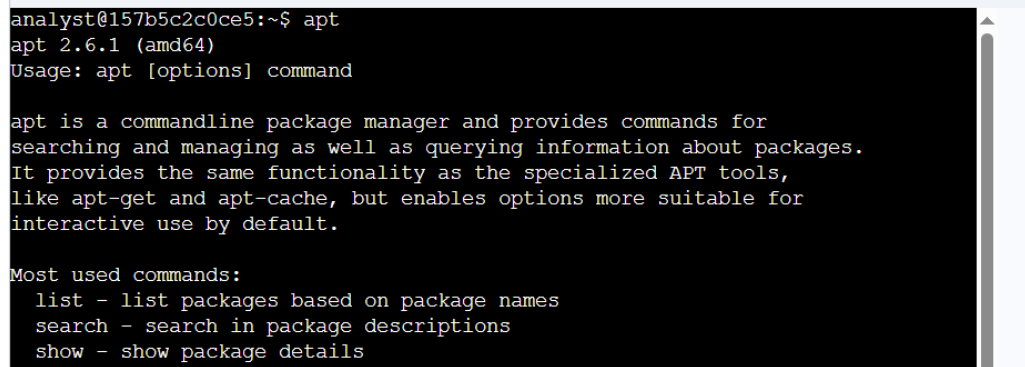
I also became familiar with Nload and tcpdump, which are useful tools for working with network traffic.

This activity helped me build my Linux fundamentals and gave me practical experience that I can use in future cybersecurity labs.

Demonstration

The following screenshots demonstrate my hands-on work completed during this activity.

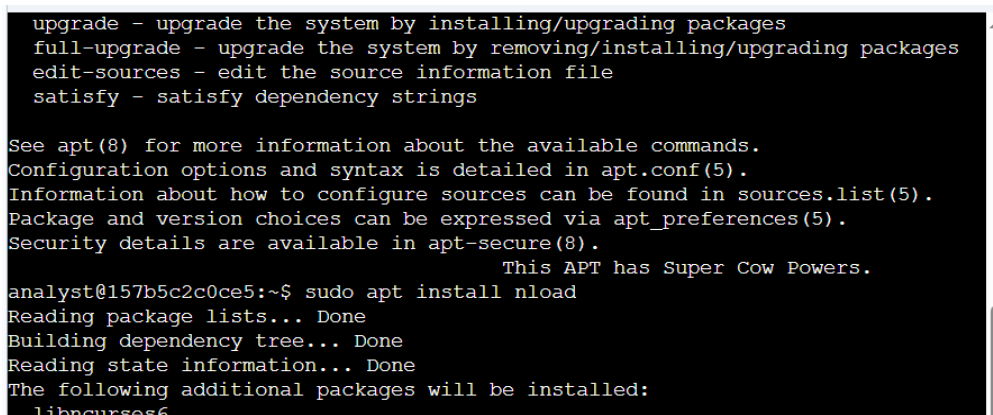
The screenshots show the Bash terminal, the APT commands I used, the installation and removal of Nload, the installation of tcpdump, and the verification of installed applications.



```
analyst@157b5c2c0ce5:~$ apt
apt 2.6.1 (amd64)
Usage: apt [options] command

apt is a commandline package manager and provides commands for
searching and managing as well as querying information about packages.
It provides the same functionality as the specialized APT tools,
like apt-get and apt-cache, but enables options more suitable for
interactive use by default.

Most used commands:
  list - list packages based on package names
  search - search in package descriptions
  show - show package details
```



```
upgrade - upgrade the system by installing/upgrading packages
full-upgrade - upgrade the system by removing/installing/upgrading packages
edit-sources - edit the source information file
satisfy - satisfy dependency strings

See apt(8) for more information about the available commands.
Configuration options and syntax is detailed in apt.conf(5).
Information about how to configure sources can be found in sources.list(5).
Package and version choices can be expressed via apt_preferences(5).
Security details are available in apt-secure(8).
This APT has Super Cow Powers.

analyst@157b5c2c0ce5:~$ sudo apt install nload
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libncurses6
```

```

analyst@157b5c2c0ce5:~$ nload -h

nload version 0.7.4
Copyright (C) 2001 - 2012 by Roland Riegel <feedback@roland-riegel.de>
nload comes with ABSOLUTELY NO WARRANTY. This is free software, and you are
welcome to redistribute it under certain conditions. For more details see the
GNU General Public License Version 2 (http://www.gnu.org/copyleft/gpl.html).

Command line syntax:
nload [options] [devices]
nload --help|-h

Options:
-a period          Sets the length in seconds of the time window for average
                   calculation.
                   Default is 300.
-i max_scaling     Specifies the 100% mark in kBit/s of the graph indicating the
                   incoming bandwidth usage. Ignored if max_scaling is 0 or the
                   switch -m is given.
                   Default is 10240.
-m                Show multiple devices at a time; no traffic graphs.
-o max_scaling     Same as -i but for the graph indicating the outgoing bandwidth
                   usage.
                   Default is 10240.
-t interval        Determines the refresh interval of the display in milliseconds.
                   Default is 500.
-u h|b|k|ml|g     Sets the type of unit used for the display of traffic numbers.

```

```

analyst@08399a8ced8a:~$ nload -h
-bash: /usr/bin/nload: No such file or directory
analyst@08399a8ced8a:~$

```

```

analyst@08399a8ced8a:~$ sudo apt install tcpdump
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libncurses6
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  libpcap0.8
Suggested packages:
  apparmor
The following NEW packages will be installed:
  libpcap0.8 tcpdump
0 upgraded, 2 newly installed, 0 to remove and 11 not upgraded.
Need to get 624 kB of archives.
After this operation, 1741 kB of additional disk space will be used.
Do you want to continue? [Y/n]

```

```

analyst@08399a8ced8a:~$ sudo apt install nload
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  nload
0 upgraded, 1 newly installed, 0 to remove and 11 not upgraded.
Need to get 56.2 kB of archives.
After this operation, 164 kB of additional disk space will be used.
Get:1 http://deb.debian.org/debian bookworm/main amd64 nload amd64 0.7.4-2+b1 [56.2 kB]
Fetched 56.2 kB in 0s (1563 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package nload.
(Reading database ... 24195 files and directories currently installed.)
Preparing to unpack .../nload_0.7.4-2+b1_amd64.deb ...
Unpacking nload (0.7.4-2+b1) ...
Setting up nload (0.7.4-2+b1) ...
Processing triggers for man-db (2.11.2-2) ...
analyst@08399a8ced8a:~$

```

The screenshots demonstrate my practical work with:

- APT
- sudo
- Nload
- tcpdump
- Installing software
- Removing software
- Listing installed applications
- Reinstalling software

Conclusion

This activity helped me understand how software is managed in a Linux environment using the APT package manager.

I practiced installing, uninstalling, checking, and reinstalling applications from the Bash shell.

I also learned about Nload and tcpdump and their connection to network monitoring and network traffic analysis.

These skills are useful foundations for my cybersecurity learning, especially as I continue working with Linux, networking, and security tools.

Course Information

Google Cybersecurity Certificate

Course: Tools of the Trade: Linux and SQL

Activity: Install Software in a Linux Distribution

Disclaimer

This document is part of my personal cybersecurity learning portfolio. The activity was completed for educational purposes as part of the Google Cybersecurity Certificate.

The lab was performed in a provided Linux virtual environment and the screenshots included in this document represent my practical lab work.